A Design and Implementation of Bangla Next Word Predictor Based on Personalized Federated Learning Leveraging Model Agnostic Meta Learning and Semantic Analysis

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***Abstract*—Next word prediction for technological devices has become a contentious issue in predictive analytics. Word prediction requires a huge amount of data from a vast number of different users, which introduces issues like heterogeneous dataset management and privacy concerns, thus raising the importance of Federated Learning (FL) as it can learn from a large amount of sparsely distributed data while ensuring privacy. Further investigations on related studies have led us to contribute in this area based on the Bangla language due to the minimal appearance of this language in the domain of word prediction based on FL. In consideration of implementing our work, some of the prime issues have been noted where avoiding data poisoning attacks, prioritizing user preferences during the global update, and focusing on fast convergence to reach a high-quality model in a few training rounds have become unavoidable matters of contention. Besides this, generating initial models which can be easily adapted by the current or new users is also an important research problem. Based on these, we have proposed a structure that combines semantic analysis and personalized FL using Model-Agnostic Meta-Learning (MAML) to obtain the desired goals. Extensive evaluations have been done on Non-IID data for both FL and personalized FL with MAML on the next word prediction model, which justifiably achieved 83.05% and 86.23% test accuracy respectively on the server model after ten communication rounds across ten clients. Apart from that, the effective results of semantic approaches show an impressive outcome as intended. Thus, the limitations and the challenges of FL on next word prediction conceptually allowed us to use a variety of algorithms to turn them into advanced ones for further usage.**

***Index Terms*—Next Word Prediction, Federated Learning, Personalized Federated Learning, Model Agnostic Meta Learning, Latent Semantic Analysis**

I. Introduction

In this century, while communicating through electronic mediums, we should make sensible use of words in order to express ourselves. Therefore, the suggestion of words can be beneficial to the users as word prediction can help us not only to write more fluently but also to generate more options to write from, provide aural assistance to evaluate word selections, reduce typing errors, and extend the user's vocabulary. In the meantime, despite the fact that 234 million people use Bangla as their primary form of communication, the language has limited work on next-word prediction. Therefore, a corpus based on the Bangla language was chosen for this study in order to develop a superior technological model for predicting the next word.

For proper next word prediction, a massive training dataset is often required, which tends to be hard to come by. Many users may not be willing to share their data because of the fear of disclosing their personal information. Additionally, processing a huge central database necessitates a significant amount of storage, which may put a heavy load on the server. Gathering a vast amount of data on a central server can also be troublesome, as many users may not be willing to share their data due to security concerns. Hereafter, Federated Learning (FL) comes up with a solution that has played a vital role in maintaining data security, which is among the most pressing matters in today's digital world. As one of the machine learning approaches, FL contributes to the confidentiality of information, data access controls, and heterogeneous data access [20]. Simply said, FL allows clients to train models without being forced to share training data with the centralized server.

Subsequently, individual users may have a unique pattern of communication. People may prefer using specific words and specific writing styles to communicate. Traditional FL aggregates contributions from all the users, forming a generic global learning pattern that often excludes minor individual patterns of communication. Precisely, users may discover that their individual preferences are not being recommended when using the model to predict the next word, as it only develops a common output without prioritizing inputs from individual users. In this instance, FL can solve this problem with the help of the Model-Agnostic-Meta-Learning (MAML) framework to create an initially distributed model. By conducting a few iterations of gradient descent on their personal data, this model can be rapidly adopted by current or new users' local datasets [7]. Because the aggregation of the local models in the server ends, the global model will also be affected.

Afterward, thinking about erring as a human has led us to severe concerns like typing mistakes, which is a common

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problem in the case of digital communication. If a client intentionally or unintentionally makes mistakes on a regular basis, the model would end up learning from them. Thus, the whole process can be discouraged in the long term. To avoid more of these kinds of data poisoning issues and to preserve the correctness of next word prediction, implementing semantic analysis (it helps to identify the semantic structures of a sentence, so those odd elements can be identified [3]) can help to check whether the client is typing something related to the topic or something unusual that's irrelevant and should be filtered and avoided from the training dataset.

The paper is organized in a manner so that it details the applications of using federated learning for ensuring data privacy in next-word prediction at the beginning. Then, semantic analysis and MAML are discussed, which can help to avoid data poisoning attacks and prioritize individuals' choices. The contributions of this research work include:

- The next word prediction in Bangla has been implemented as there is a lack of work on this topic in the Bangla language.
- The application of FL addresses privacy concerns and points out some of the problems caused by heterogeneous dataset management issues and is working on their solutions. For example:
  - Reducing the effect of unwanted text or malicious data in global updates. We are using a semantic approach in the client-end model to address this.
  - Then, to personalize the client end model by prioritizing local updates compared to the global one, initializing shared models that can be easily adapted by all users, and converging the models rapidly for better performance in short training rounds. For this, we have applied personalized FL with MAML.

II. Literature Review

This section will critically evaluate prior noteworthy studies in the area of word prediction in the Bangla language and the utilization of personalized FL with MAML alongside LSA. To begin with, an LSTM model was used to construct an accurate next-word prediction model in [22]. There has been some similar next-word prediction work on the Bangla language based on different models; for example, LSTM in [11], Stochastic language model in [9], and Gated Recurrent Unit (GRU) in [18] were applied on their collected dataset. In [2], it was illustrated that LSTMs had improved the decoding of gestured inputs on mobile keyboards with the combination of conventional Finite State Transducers. Various techniques are used to improve user experiences through the use of ML. For example, in [10], better updates to the features of Gboard were brought with a huge heterogeneous dataset, maintaining privacy concerns. Then, in [21], by improving query clickthrough rate (CTR) while taking suggestions from the baseline model and removing low-quality suggestions through the triggering model.

In the research [19], in addition to noting many possible vulnerabilities due to data poisoning assaults in FL and the impact of malevolent participants, some defense techniques were suggested, such as clustering model gradients. [3] claimed that identifying connections between text components and performing the latent semantic analysis (LSA) may accurately estimate users' preferences. Moreover, according to [12], malicious data was accurately filtered using the LSA and MD5 algorithms, which outperforms other filtering algorithms such as SVM and naive Bayes. In parallel, according to another paper, personalizing FL with respect to MAML methods can not only ensure all the benefits of FL but can also generate a far better, easily adaptable model for each user [5]. Additionally, it captures the differences of all users, which can be used as a solution in preliminary steps and help to improve models with their personal data [6]. Furthermore, the MAML technique in convergence with FL is the process of nodes executing numerous updates locally and then submitting them to the server, which has not been explored often in the field of meta-learning [14].

There have been some works that suggest multi-task training methods, which also lead to a collection of user-centric solutions. The implementation of FL with respect to meta-learning, more specifically multitasking, has been described in [13] & [15]. In [8], the authors explored a training structure for a combined package of a global model and localized models, which culminated in a customized approach for each user. An adaptable FL algorithm that combines local and global models is represented as the personalized model in [4]. Hereafter, in terms of improving algorithms using differential privacy methods and SGD, it was finally discussed in [17], [23] and [1].

III. Methodology

We would like to describe the methods that have been applied to achieve our predetermined research objectives.

A. Bangla Next Word Prediction

As illustrated in Figure 1, we primarily preprocessed the Bangla Corpus dataset by removing all kinds of emoticons, symbols, and Latin letters (a-z, A-Z, 0-9, etc.). We kept the punctuation marks as the following words to predict. Afterwards, the model created tokens which resulted in 58573 individualized words in the dataset. Later, the entire collection of sentences was divided into a sequence of length 5, and the sequences were encoded accordingly. Then, at a time, a dictionary was composed where 8626 unique words were stored, and input-output was initialized. As a result, a sequence of length four was formed as input, and a binary matrix representation was produced from the following sequence. Before moving forward, we split the inputs and outputs into an 8:2 ratio for training and testing, and then Bidirectional LSTM (Bi-LSTM) was applied, which can perform well to preserve knowledge from the previous and upcoming context at any given point. The Bi-LSTM Neural Network has three different types of layers, which are the embedding layer, the LSTM layer, and the linear layer. Here, we have some hyperparameters, which include the number of epochs, which

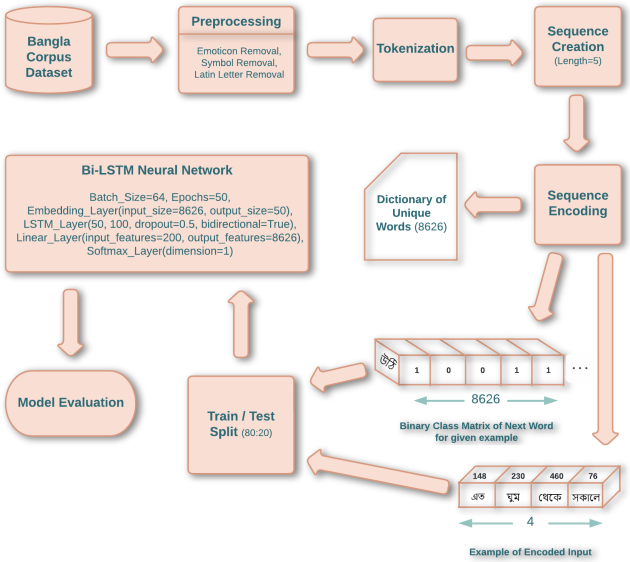

Fig. 1: Flowchart of Next Word Prediction

was set to 50, along with 50% dropout, and a batch size of 64. Finally, the evaluation of the trained model was conducted to measure performance.

B. Problem Formulation

In this subsection, we demonstrate the problems that we identified during our research. Based on these research questions, our paper has moved forward toward its goal. All the research questions (RQ) are given below.

- RQ1: Privacy concerns about the massive dataset
  - How to store a huge dataset and manage it on a central server?
  - In terms of decentralized training, what is the most efficient way to handle a large dataset with a large number of clients?
  - How can we provide maximum security to those clients' datasets?
- RQ2: Data Poisoning Attack from Malicious Participants in FL
  - How does it affect the global model if unwanted or inappropriate data is shared from the client end?
  - What should be the precaution against poisonous data from unethical clients?
- RQ3: Prioritize adaptation of heterogeneous dataset in local model
  - How do we prioritize the local updates to personalize the client model besides the global one?
  - In what way should the models be updated so that current or new users can easily adapt?
  - How to ensure fast convergence of the model during training rounds?

Further discussions of our paper are based on these formulated problems and are designed according to Figure 2, which demonstrates our further work procedure.

C. Federated Learning

A solution for RQ1 in III-B has been explored in this subsection. Due to storage constraints and leakage of sensitive personal data, our prediction model, defined as the global model, is initially transferred to selected client devices. After completing the training session on idle time, client devices will send their trained model back to the server. More simply, our model follows the traditional workflow of FL developed by Google in 2017 [16]. The server then uses the FederatedAverage (FedAvg) algorithm to aggregate each weight obtained from the clients, modifying the global model as an outcome.

$$\sum_{c=1}^{C} \frac{u_c}{N} w_{r+1}^{c} \rightarrow w_{r+1} \tag{1}$$

Here in (1), we have denoted r, as communication round, w_r as a global model, C as a subset of client devices, u as the sum of all users and u_c samples are in each of clients' local dataset per round. Note that, w_r is sent to C at r, each u_c varies from device to device and $N = \sum_c u_c$. Then, a weighted aggregation of client models is conducted in the server to obtain w_{r+1} which is a new global model.

$$w_r - \epsilon g_{avg} \rightarrow w_{r+1} \tag{2}$$

Now, as per (2) using one or more steps of stochastic gradient descent (SGD), each client computes g_{avg} the average gradient. For a client learning rate, the clients are locally completing SGD updates on their local dataset, sending them to the server for aggregation, and updating w_{r+1}. Furthermore, models need to learn about the limitation that device-generated data is saved and processed locally, with only intermediate changes transferred to a central server on a regular basis. In particular, to achieve the goal of minimizing the function in (3),

$$\min_{w \in \mathbb{R}^d} f(w), where f(w) = \frac{1}{u} \sum_{i=1}^{u} f_i(w) \tag{3}$$

$f_i(w)$ represents the resulting loss where each model w, generates a $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$ which represents i user's anticipated loss across shared data.

In our case, in the first round, ten client devices got the same global model pre-trained on proxy data representing the primary Bangla corpus dataset. Then, after processing updates in the next ten rounds in the local memory, the client sent the generated weight vector for aggregation to the server to improve the global model and clear their local memory. It is imperative to note that while user data is unreachable from the server end, clients must be confident that the aggregation will not evaluate specific weight transferred in order to use the FL approach described here. Therefore, RQ1 can be solved as data is non-shareable with the central server and is managed on

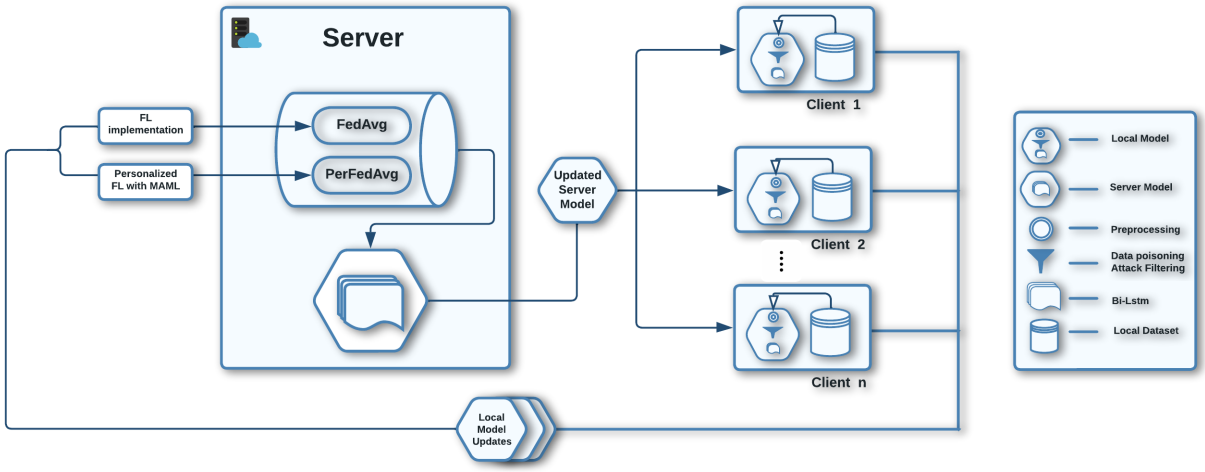

Fig. 2: Design of FL and Personalized FL

the client's end; FL ensures maximum security for the client's dataset.

D. Semantic Analysis

RQ2 of III-B has been emphasized in this section of the work, discussing a problem of the real-life FL approach briefly. Due to privacy constraints, the non-shareable dataset remains at the client's end; the server is unable to influence this dataset or determine whether the clients are using appropriate or inappropriate data for training local models. Thus, during the time of computing FedAvg, the server gets affected by all the local data inputs as the client models are trained locally. Additionally, due to the nature of FL, the datasets are generated, stored, and updated at the client's end, where unintentionally inputting malicious data through typing mistakes is really common and intentionally attacking a model with a poisonous dataset is also quite feasible. According to our case, to protect the whole system from data poisoning attacks, precautions should be taken so that a subset of malicious FL Clients can not attack with their own datasets and hamper the global model, which is to be updated by the local models that are trained by the injurious dataset.

On the other side, meaningless or inappropriate input can be identified with the help of semantic analysis. Moving forward to Latent semantic analysis (LSA), which is a completely automated computational tool for obtaining and implying predicted contextual use of phrases in textual sequences. As predicting the following word is a representation of text sequences, LSA is beneficial as a type of semantic approach. Here, LSA has two major steps, which are: creating a document term-matrix from a raw dataset to represent texts as points in a vector and performing a singular value decomposition (SVD) on it to analyze principal components. SVD helps to reduce the dimensionality of the original dataset by encoding it using latent features.

Here, Z and Z' are the feature-document matrices composed using the vectors Y for all documents as columns, and the column space of Z' is a subspace of the column space of Z. Technically, based on the SVD of Z, a low-rank approximation Z_k replaces the original Z by Z', were in (4),

$$Z = X \sum Y^T \approx X_k \sum_k Y_k^T =: Z_k, \tag{4}$$

Here, the k largest singular value of Z is contained in $\sum_k$ where, $k < f$. In our case, right between completing the preprocess and training the model, LSA will be conducted on each client's dataset, which is shown in Figure 2. Then, using the results of the analysis, the inappropriate inputs can be marked and removed from the training dataset. As the dataset is being cleaned even before using it for local training, during the aggregation of all the local models, the global model in the server can not be harmed. Thus, by cleaning up the training data using semantic analysis, the server presumably stays protected from unethical attacks and thus resolves RQ2.

E. Personalized FL with MAML

The foundation of federated learning has some flaws, as RQ3 in III-B points out. To solve these issues, we introduce the idea of personalized federated learning via model-agnostic meta-learning [6]. FL is essentially concerned about the maximum security of clients' datasets; on the other hand, it does not address some client-side difficulties, such as the client often desires for their prediction to be prioritized based on their own training sample rather than the global model's prediction. While using personalized FL with MAML in our proposed next-word prediction scenario, we seek a model initialization point that can be updated based on the client's own dataset through a few gradient descent rounds to build a well-performed prioritized prediction model as desired.

Moreover, in (5) Personalized FederatedAverage (perFedAvg), like FedAvg, delivers the proposed global model w_r

to a portion of clients U_r chosen randomly with size rz, during round k. Each user $u \epsilon U_r$ conducts SGD for τ steps locally with respect to the meta-function M_u associated with user u. Here, the local update sequence for $S \geq s \geq 1$,

$$w^u_{r+1,s} = w^u_{r+1,s-1} - \beta \tilde{\nabla} M_u(w^u_{r+1,s-1}) \quad (5)$$

Here, the learning rate of local training represented by β and $\nabla M_u(w^u_{r+1,s-1})$ indicated gradient function which is the form of,

$$\nabla M_u(w) = (I - \alpha \nabla^2 m_u(w)) \nabla m_u(w - \alpha \nabla m_u(w)) \quad (6)$$

Now, inspired by (6),representing D^u as a batch of data and α as a learning rate in (7), $\tilde{\nabla} M_u(v) =$

$$(I - \alpha \tilde{\nabla}^2 m_u(v, d^{''})) \tilde{\nabla} m_u(v - \alpha \tilde{\nabla} m_u(v, d), d^{'}) \quad (7)$$

Thus, computing the stochastic gradient $\nabla M_u(v)$ where $v = w^u_{r+1,s-1}$, for all local iterates using batches $D^u_s = d$, $D^{'u}_s = d^{'}$ and $D^{''u}_s = d^{''}$, it generates the local sequence $\{w^u_{r+1,s-1}\}^{\tau}_{s=0}$ which finally leads us to a personalized local update.

Henceforth, all of the local models are passed on to the server for subsequent processing. Afterwards, the server modifies its model by averaging them, as seen in (8),

$$w_{r+1} = \frac{1}{rz} \sum_{u \epsilon U_r} w^u_{r+1,\tau} \quad (8)$$

Hereafter, this updated global model is transferred to randomly selected clients due to the continuance of the following cycle.

This technique actually resolves the other problems that arose in RQ3. The above-mentioned updated personalized FL model not only keeps the benefits of FL by managing and securing a vast dataset of clients but also provides more adaptable, fast convergent features according to the heterogeneity of individual clients' datasets. As we discussed in the equations, the models ensure rapid convergence with the help of SGD during the training rounds. Additionally, every selected client gets the global model, which can easily adapt to the new client after a small number of iterations through their dataset. Thus, this easy adaptation also enables the model to behave according to the client's priority based on their own dataset besides the global shared one.

IV. Experiment

This section indicates the experiments that we have done for evaluation during our research.

A. Dataset Formulation

Owing to the unavailability of a standard Bangla dataset, we formulated a dataset that can represent a standard Bangla corpus during the process of our research. The sentences were gathered from a variety of newspaper passages and comment sections of Bangla YouTube channels. They included positive, negative, emotional, aggressive, and other combinations of sentences for the dataset to voluntarily reflect the genuine diverse Bangla corpus. As a result, the cumulative amount of words gathered in our dataset is around 58763. Here, Figure 3 shows our dataset's word cloud, which includes some random words from our Bangla corpus. After that, all of the collected data was combined and saved as a .txt file for onward usage.

Fig. 3: Bangla Word Cloud

B. Evaluation

The accuracy metric, which is the percentage of predictions our model correctly predicts, is applied to assess how effectively the algorithms perform. Figure 4 illustrates the differences in accuracies of different algorithms, including Bangla next word prediction with a default model, FL model, and personalized FL model. Moreover, comparing the server-side accuracy of FL and personalized FL with the default model can provide us with a cumulative view of the algorithm's performance. From the observations made in this figure, the highest accuracy (86.23%) obtained by the algorithm is personalized FL. However, this algorithm surpasses both of the other algorithms, where FL scores 83.05% and the Bangla next word prediction's default model acquires 81.65% accuracy. Based on these findings, we determined that personalized FL is the optimal model.

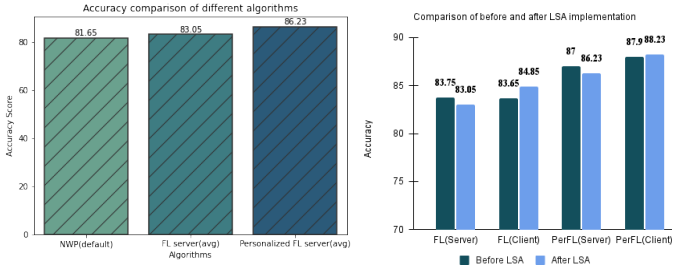

Fig. 4: Comparison of Accuracies between Algorithm specifying Before and After Implementation of LSA

Afterward, we used LSA to analyze the before-and-after effects of our above-mentioned algorithms in III-C and III-E, which are shown in Figure 4, in order to evaluate the effectiveness of data poisoning attack prevention in the FL environment. First, we directly injected three unethical clients with various malicious data into our FL environment. The accuracy is not decreased after malicious data is placed, as

can be shown in Figure 4 as the semantic analyzer filters out all of the malicious data on the client side. In the FL setting, the server experiences a kind of equal performance (83.75% vs 83.05%), where the clients get slightly better accuracy after implementing LSA (83.65% vs 84.85%). In a personalized FL environment, the server accuracy slightly decreased (87% versus 86.23%) while also marginally improving on the client side (87.98% vs 88.23%). In conclusion, it is clear that the implementation of LSA, which generates the desired enhanced authenticated model, allows it so that data poisoning attacks do not negatively impact the client and server side.

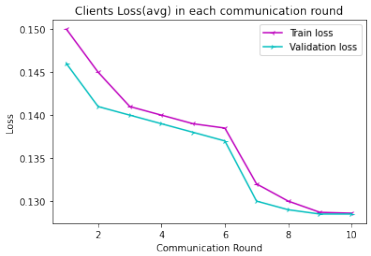

Fig. 5: Average of Client Loss graph

Additionally, Figure 5 shows the average client loss in each communication round of our recommended personalized model to analyze performance improvement. After each communication round, we notice that the difference between the training and validation curves has been minimized, indicating that our model seems almost optimal. Moreover, the loss graph shows a downward tendency throughout the communication rounds. These criteria actually demonstrate the model's performance improvement throughout the communication rounds with a gradual decrease in error.

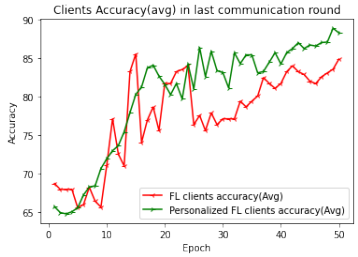

Fig. 6: Client Accuracy graph of FL and Personalised FL

Furthermore, it is important to evaluate how the clients performed during the communication rounds. For the performance demonstration, we gathered the average client accuracy value for each epoch during the last communication round (for both FL and personalized FL), as shown in Figure 6. It is rather obvious that both algorithms' average client accuracy rises, which reveals that both models' performance gradually gets better. However, personalized FL outperformed FL in terms of average client accuracy after 50 epochs, with an accuracy of 88.23 compared to 84.85.

On the other hand, we examined personalized FL's effect on the client side, with results shown in Figure 7. In this figure,

Fig. 7: Example of next word prediction using Personalised FL with MAML

we illustrated an example of our work in which ten clients participated. In this instance, nine out of ten clients texted "Extremely thanks for inviting us" (translated from Bangla). In contrast, client number one texted "Extremely thanks for coming" (translated from Bangla) across all communication rounds. After the completion of training, we input 'Extremely thanks for' (translated from Bangla) to examine what word would display next on each client device. Except for client 1, all other clients get the next word 'inviting', as the server should provide it to all the clients according to the rules of FedAvg. However, client 1 receives the next word 'coming' after some training with respect to its own data. This different appearance essentially denotes the prioritization of the client's model over the global model and the fast convergence of the model according to the client's data during the training round, as we desired.

Finally, it is obvious to come to the conclusion that personalized FL outperformed all other algorithms according to its promising performance. The final structure of personalized FL leveraging MAML and LSA has proven itself as the best answer to all of the individual research questions from III-B.

V. Conclusion and Future Works

Conclusively, our proposed structure combines FL, MAML, and semantic analysis for the betterment of devices' learning processes on how to predict the next word in Bangla, which contributes to several aspects. Specifically, we have structured an advanced model to ensure massive data security and management, avoid intentional or unintentional data poisoning attacks, prioritize clients' preferences based on the heterogeneity of data, and create an easily adaptable model for new and existing clients. Concerning our experimental results, we successfully structured the improved model in a way that is beneficial in real-life implementation. As an extension of our work, a sentiment-based prediction of the next word can be employed in FL scenarios to increase client usability since an individual's diversity of sentiment can impact the choice of the next words. Along with this, an impartial FL system with a global model capable of calibrating the size of distribution based on each client's computational capacity and preventing untargeted poisoning and backdoor attacks at the client end could be a significant contribution. Finally, this research has the potential to have a significant impact on our daily communication systems via improved FL.

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